# Utilizing Convolutional Neural Networks for Brain Tumor Segmentation in MRIs

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## Abstract

Brain tumor segmentation in magnetic resonance imaging (MRI) is the gold standard for the accurate diagnosis, monitoring, treatment of gliomas, the most common and aggressive type of brain tumor. Despite its importance, much of brain MRI segmentation is performed manually, a task that is laborious, prone to human error, and features low reproducibility due to high variability in tumor structure and presence of imaging artifacts. Automated and semi-automated segmentation using machine learning yields promise in assisting clinicians in faster tumor identification and localization. In this project, we implement a convolutional neural network (CNN)-based segmentation algorithm built using a U-Net architecture to perform binary whole-tumor segmentation on multi-modal MRI data from the BraTS2020 dataset. Expanding on the work of Pereira et al., our primary objective was to evaluate model image segmentation performance under the following preprocessing techniques: baseline (no preprocessing), Gaussian smoothing, and intensity normalization. To optimize model performance, MRI volumes were grouped by subject to prevent data leakage, converted into 2D axial slices, and joined into four-channel inputs corresponding to T1, T1Gd, T2, and FLAIR imaging modalities. Model performance was evaluated using the Dice Similarity Coefficient (DSC) , Positive Predictive Value (PPV), and Sensitivity. The results demonstrated that the CNN-based approach significantly outperformed traditional intensity-based thresholding method, which failed to produce any meaningful segmentation. All preprocessing techniques yielded strong performances (DSC > 0.80 mean), with intensity normalization achieving highest tumor pixel overlap, and Gaussian smoothing improving overall precision. Despite high average performances, variability was observed across individual slices, including cases of complete segmentation failure. These findings indicate that while preprocessing techniques can improve specific performance metrics, no single method can universally optimize all outcomes. Overall, this study reinforces the effectiveness of deep learning-based approaches to augment MRI segmentation while highlighting the significance of dataset size used for training, effects of preprocessing strategies, and proper selection of model training hyperparameters in achieving consistent and reliable results.

## Introduction

Brain tumors are an abnormal, uncontrolled growth of cells within the brain or surrounding structures that can result in both physical and behavioral changes. Among them, gliomas are the most common and aggressive formation of brain tumors, leading to a shortened life expectancy and often associated with substandard prognosis in the high-grade cases [Pereira et al, Morosan et al]. As gliomas exhibit the highest mortality rate, treatment at early stages is critical to the improvement of quality of life. This serious health condition requires extensive imaging and accurate medical image processing to help locate and treat the growth. Precisely locating the tumor is imperative to the administration of non-invasive treatment options to decrease risk, promote faster recovery, and preserve healthy brain tissue. [Bauer *et al*].

Magnetic resonance imaging is the conventional imaging method to detect, diagnose, and monitor brain tumors in clinical settings. Brain MRIs can offer high resolution images of the brain, blood vessels, and surrounding soft tissues when used in conjunction with contrast agents like gadolinium. These contrast agents serve to alter the magnetic properties of water molecules near these structures, which will provide the vital information to then assess from the images. Clinicians use MRI to assess and locate brain tumors; however, the large amount of data produced by MRI impedes the use of precise quantitative measurement systems for timely diagnostics. To support the adoption of MRI into clinical practice, reliable and automatic segmentation is required. However, automatic segmentation is challenging due to large spatial and structural variability among brain tumors.

Magnetic resonance images are often imperfect and will contain noise and other image artifacts, thus potentially benefitting from image processing techniques such as intensity-based normalization and Gaussian filtering. Methods such as these aid MRI scan segmentation by reducing noise and artifacts, allowing for downstream segmentation models like Convolutional Neural Networks (CNNs) to converge faster and with higher accuracy. Advances in these neural network based approaches have been useful in tumor segmentation. Pereira et al. proposed their CNN-based method for brain tumor segmentation utilizing multi-modal data, and found that implementing their preprocessing techniques displayed improvement in performance of the CNN predictions. Through improved understanding of spatial and intensity-based features, their CNN-model was able to provide high performance metrics. Furthermore, their average running time was 8 minutes, taking only 10% of the time when compared to other methods. Building off of this work, we propose a similar implementation of a CNN-model with 3x3 convolutional kernels.

In this project, we implement a CNN-based segmentation utilizing a simple U-Net architecture applied to the multi-modal MRI slices. This is focusing on the binary whole-tumor segmentation specifically, where the CNN model will predict a single mask that represents the entire tumor

region. Alongside the convolutional neural network, a very simple thresholding based segmentation method will be implemented to show an initial baseline for further comparisons. We will assess the different preprocessing methods, including baseline (no preprocessing), Gaussian smoothing, and intensity normalization to determine the impact of these strategies. The runtime and prediction results will be compared to those obtained by Pereira et al.

## Methods

### I. Methodology

The workflow consists of various steps to load and prepare the data for further evaluation by the models. All of the following steps took place; data loading, organization by volume, dataset splitting, 2D slice extraction, preprocessing, model training, and evaluation. This structured pipeline was done to verify that the segmentation is computationally efficient for our limited hardware and consistent across the different preprocessing technique implementations.

The dataset first was grouped by volume levels before any further extraction of slices was performed. This was done to confirm that the subject's specific data stays grouped together during training and evaluation, to prevent any random slices from interfering with the results. Following this, the dataset grouped by volumes was then split into training, validation, and testing subsets. This approach is vital in order to prevent leakage of data or artificially inflated performances. Each of the volumes contain many different 2D slices, so the structures could potentially appear in both the training and testing sets if you do not split them up at the volume level.

After splitting, the MRI volumes are converted into 2D axial slices to serve as model inputs. A slice-based method is utilized to reduce the amount of computational complexity. Each MRI slice contains four modalities which are merged to form a four-channel model input. These modalities contain complimentary information that allows the CNN to learn the fine features that are prerequisites to proper learning.

The ground-truth masks provided in the dataset cover a variety of different tumor subregions. For consistency across techniques, these segmentation masks were united in a single binary mask that represents the whole tumor region. This serves the role of helping provide alignment across the model's visual outputs and evaluation metrics.

Three strategies for preprocessing were evaluated: Gaussian smoothing, Intensity Normalization, and Baseline (No preprocessing). The preprocessing was implemented prior and independently to each MRI channel before the channels were then merged into the multi-channel input. These specific preprocessing techniques were chosen to determine the impact of normalization on

standardizing the distribution of pixel intensities & Gaussian smoothing for reducing the noise and artifact quantity.

To implement segmentation, a U-Net CNN architecture was utilized. These convolutional layers contain 3x3 kernels, which are small enough to be computationally efficient but effective enough to learn the spatial features desired. The model outputs a map that shows the likelihood of each pixel appearing to be a part of the tumor. The values obtained from this are converted into a binary mask through setting a threshold, where the pixels above the threshold are identified as part of the tumor and the rest are classified as background.

Alongside the CNN model implemented, a very simple threshold-based segmentation method was included as a baseline to compare. This method utilizes thresholding through intensity levels and generates binary masks, and can provide some outlook on how the CNN approach performs.

The complete workflow of the CNN segmentation pipeline is condensed in Figure 1.

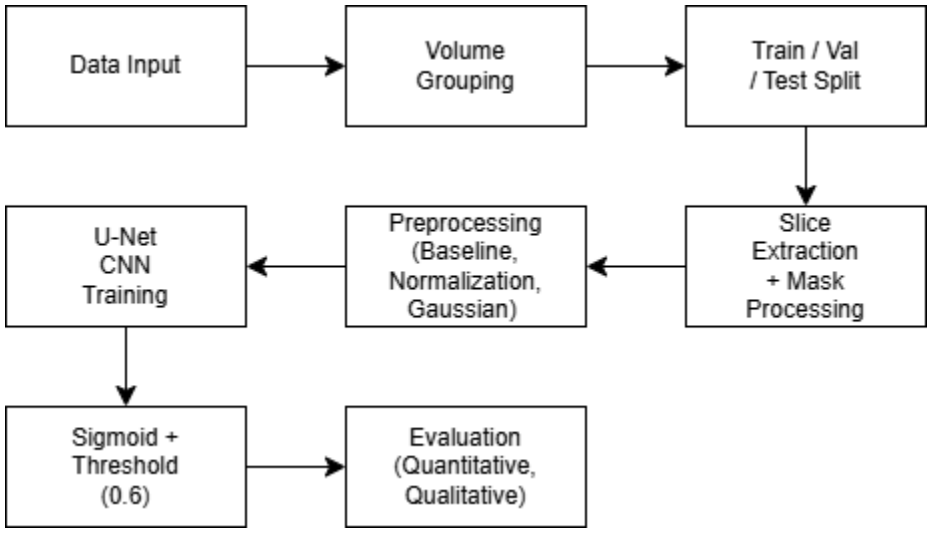



**Fig 1.** General workflow of the CNN segmentation pipeline. Data is loaded, then grouped by volume to prevent data leakage, followed by the training, validation, and testing splits. Slice extraction was done before implementing preprocessing techniques and subsequently the U-Net training and output processing are performed before final evaluation.

## II. Data Description

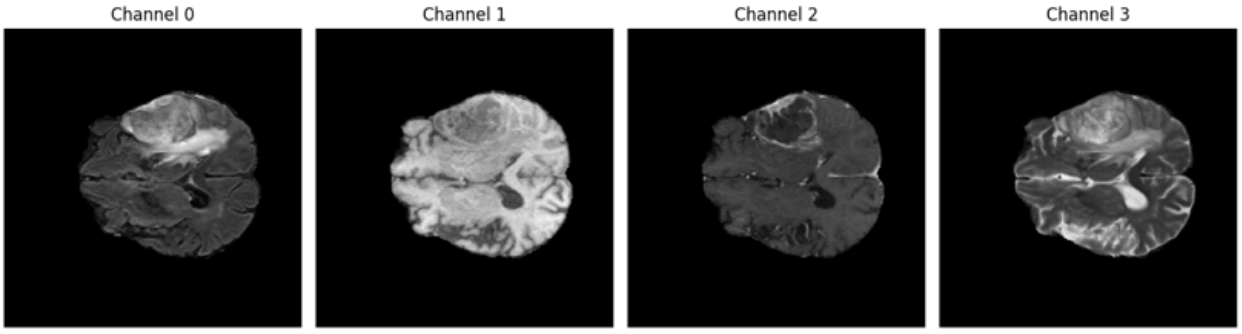
This project uses the Brain Tumor Segmentation dataset (BraTS2020) from Kaggle, which is a publicly available and multi-modal MRI dataset containing scans alongside ground truth annotations from field experts. The inclusion of these ground truth annotations provides an

excellent basis to assess the performance of segmentation alongside the other validation metrics that were utilized.

Each of the subjects within the dataset contains preprocessed modalities (4 for each subject), signaling that all of the images included are spatially aligned and contain the same dimensions. The native (T1) modality provides insights on the anatomical detail and the general structure of the brain tissue. Post-contrast weighted T1 (T1Gd) focuses more on accentuating the contrast-enhancing tumor regions. Finally, the T2-weighted (T2) and T2 Fluid Attenuated Inversion Recovery (T2-FLAIR) volumes are effective at highlighting visually the edema and various other abnormalities.

This dataset contains multi-modal 3D volumes with dimensions of 240 x 240 x 155 voxels. It also contains a voxel resolution of $1mm^3$. This kaggle dataset contains files that were stored in .h5 format (HDF5). Each specific sample is one slice from the original 3D MRI volumes containing the four imaging modalities alongside the ground truth segmentation mask.

Through initial inspection of the dataset, we were able to confirm that each sample had four modalities (T1, T1Gd, T2, and FLAIR) with their ground truth segmentation mask in conjunction.



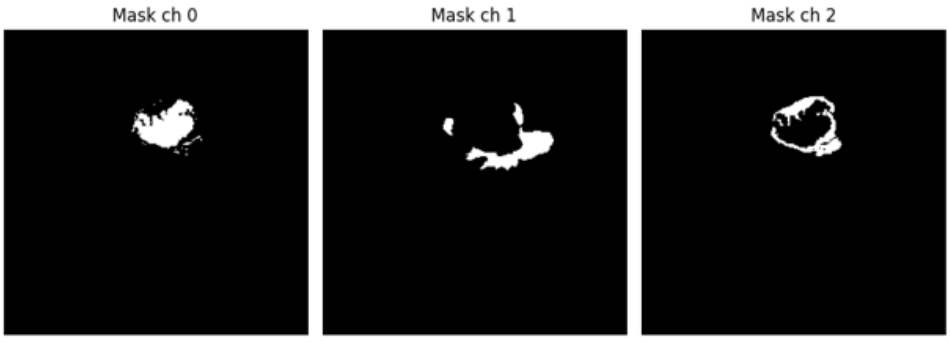



**Fig 2.** Representative multi-modal slice from the BraTS 2020 dataset showcasing the four modalities and corresponding ground truth masks. Each channel provides supplementary data upon one another for better tumor identification.

As displayed in Figure 2, the tumor can be visualized across the modalities and the ground truth mask provided.

For our project, the 3D volumes were first grouped together then split into 2D axial slices for input into the CNN and simple thresholding models. Only a small subset of volumes was utilized due to the time and hardware constraints.

## III. Validation Method

To evaluate the quality of the segmentations, there are three different metrics that we implemented. First being the Dice Similarity Coefficient (DSC).

$$DSC = \frac{2TP}{2TP + FP + FN}$$

This will measure the overlap between both manual and automatic segmentation and give us a metric of comparison.

The next metric would be Positive Predictive Value (PPV); which is a statistic examining the number of tumor pixels that have been correctly identified.

$$PPV = \frac{TP}{TP+FP}$$

Sensitivity will be another metric that can measure the ability to identify true positives (identification of tumor pixels in this case).

$$Sensitivity = \frac{TP}{TP+FN}$$

Our criteria for success was to look for high DSC scores alongside ideal PPV and Sensitivity metrics. Visualization of the method is also crucial for ensuring that successful segmentation has occurred. The predicted segmentation masks were directly compared to the ground truth annotations as well to determine model performance. Finally, we observed the computational time of image preprocessing and segmentation and compared it to the results of the method utilized by Pereira *et al.*

## IV. Implementation Details

This project was implemented utilizing Jupyter Notebooks written in Python. The PyTorch deep learning framework was used to integrate a U-Net convolutional neural network. The MRI data provided from the Kaggle dataset was loaded from HDF5 formatted files (.h5), with each sample containing the multi-modal slice alongside the correlative ground-truth mask.

The dataset was then organized by volume to prevent data leakage as discussed in the methodology. A subset of 5 volumes was chosen to further split into training, validation, and test sets. Ideally, we wanted to work with a much higher volume than this but due to the computational constraints and runtime of even the 5 volumes, this seemed unrealistic. After selecting the volumes, they were split into 70% (training), 15% (validation), and 15% (test) subsets. Since only 5 volumes were able to be run; this resulted in only three volumes being trained and one volume for each of the validation and testing.

After the splitting, the data was used in the 2D axial slices state. Each of the slices contained their four input channels corresponding to each modality. The segmentation mask was then transformed into a binary whole-tumor mask so that our outputs and evaluation stayed consistent.

The preprocessing techniques were implemented before the channels were combined into the four-channel input tensor used for the U-Net architecture. We compared baseline (no preprocessing involved), Gaussian smoothing, and intensity normalization. Each method was run independently to evaluate the differences between them and to determine if there is meaningful improvement when utilizing preprocessing before inputting into the convolutional neural network model. Gaussian smoothing was done independently to each MRI modality provided using a Gaussian filter with $\sigma = 1$. Intensity normalization was executed utilizing z-score normalization prior to CNN input. The baseline condition included no preliminary steps in preprocessing.

The U-Net architecture utilized was a PyTorch Hub model which was arranged for four input channels and one output channel. This model was specifically trained with the Adam optimizer and also included a Binary Cross-Entropy (BCE) and Dice loss function to try to amend any class imbalances that could be occurring within the dataset. The three preprocessing conditions were each trained with 5 epochs. In future runs we would like to train longer but due to computational constraints resulted in 5 to get results in a realistic time frame. Training and validation loss was monitored through epoch training to understand how well the model is learning. Table 1 displays the general settings that were implemented with our U-Net CNN model.

**TABLE 1.** The summary of the U-Net CNN settings that were utilized for binary whole-tumor segmentation. The PyTorch model used 4 MRI modalities as input channels with 1 binary

whole-tumor mask output, using BCE + Dice Loss functions, Adam optimizer, an 0.001 learning rate and 5 epochs. The sigmoid activation was set for output with a thresholding of 0.6.

| **Settings** | **What's Utilized** |
|---|---|
| Model Architecture | U-Net (PyTorch Hub) |
| Input Channels | 4 (T1, T1Gd, T2, FLAIR) |
| Output Channels | 1 (binary whole-tumor mask) |
| Loss Function | BCE + Dice Loss |
| Optimizer | Adam |
| Learning Rate | 0.001 |
| Epochs | 5 |
| Output Activation | Sigmoid |
| Threshold | 0.6 |
| Preprocessing Compared | Baseline, z-score normalization, Gaussian ($\sigma = 1$) |

Utilizing a sigmoid activation function and a thresholding value of 0.6, the model outputs how likely each pixel corresponds to the tumor. The pixels above this threshold were classified as a part of the tumor while the others were determined as background. 0.6 was more effective in providing desirable metrics.

In addition to implementing the CNN, we used a very simple threshold-based segmentation method without any tweaks to determine if this is an effective method of segmentation in comparison.

Model performance was done through visualizations including predicted masks as well as overlap between the ground-truth masks. Metrics including Dice Similarity Coefficient (DSC), Sensitivity, and Positive Predictive Value (PPV) were another way to determine how effectively the model performed.

While the basis of our implementation was done on the Pereira paper approach, there were several modifications that needed to be made due to both dataset and computational limitations. Instead of the patch-based approach done by their study, we implemented 2-D slice based inputs with a simplified U-Net architecture. The data trained, validated, and tested was also on a much smaller scale in comparison.

## Results

Both qualitative and quantitative results were analyzed to determine the performance of the various preprocessing methods on the CNN-based model. Some differentiation between the two implementation methods (threshold-based and U-Net) is also made, but with little to no tweaking of the threshold-based model as it served this use-case ineffectively. All of the visual comparisons are also substantiated with DSC, PPV, and Sensitivity metrics to give more credence to exactly how well they are performing.

### I. Threshold-Based Segmentation

An intensity-based segmentation method through thresholding was first utilized as a baseline for comparison with the U-Net model. Using only the intensity values, this method determines pixels as either tumor or background. This was done at a very rudimentary level and provided no benefit in segmentation of the tumors with poor metrics to substantiate this.

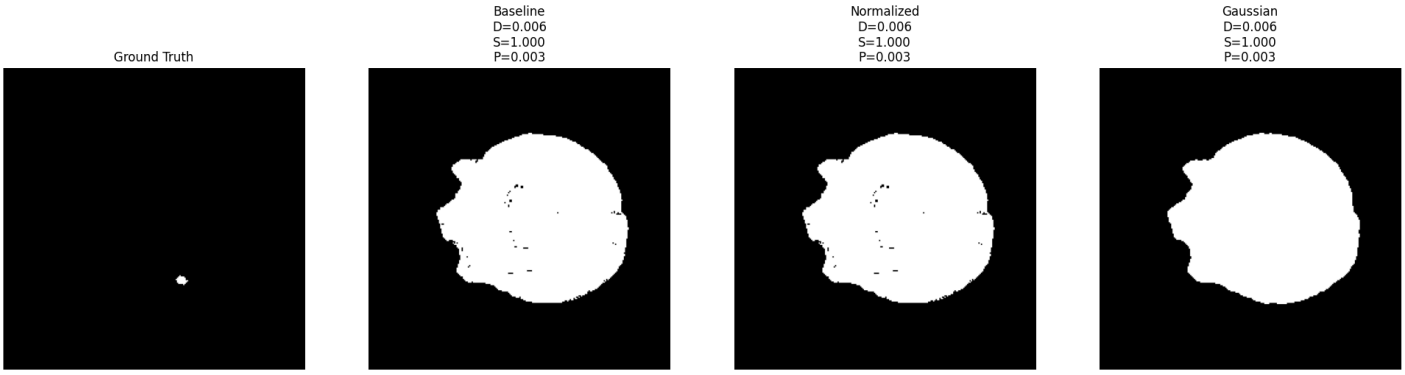



**Fig 3.** Failure of threshold-based segmentation results for baseline, normalized, and Gaussian preprocessing in comparison to the ground truth. Identical poor performing metrics displayed above the images. (DSC = 0.006, S = 1.000, PPV = 0.003).

As seen in Figure 3 above, all preprocessing methods exhibited poor overlap with the ground-truth tumor region. The identical performance statistics across these show that the threshold-based method is fundamentally constrained for this task. It is unable to understand the more subtle differences in intensity needed to properly segment. Seeing a sensitivity of 1.000 is showing that this model is oversegmenting.

### II. CNN-Based Segmentation

To follow the implementation done by Pereira, we utilized a CNN model to improve upon tumor segmentation. Unlike the previous intensity thresholding technique, the convolutional neural network was able to learn fine details and features allowing for much more comprehensive and

accurate identification of the tumor pixels. Each conditional preprocessing method was run and mean metrics computed to quantitatively understand their performances.

In order to assess the model's generalization and convergence rate, we measured the training and validation loss over each epoch. Figure 4 showcases the loss curve for the CNN-model under the baseline preprocessing condition. This was chosen due to its highest performance across the three preprocessing techniques, as it showed the highest quantitative statistics.

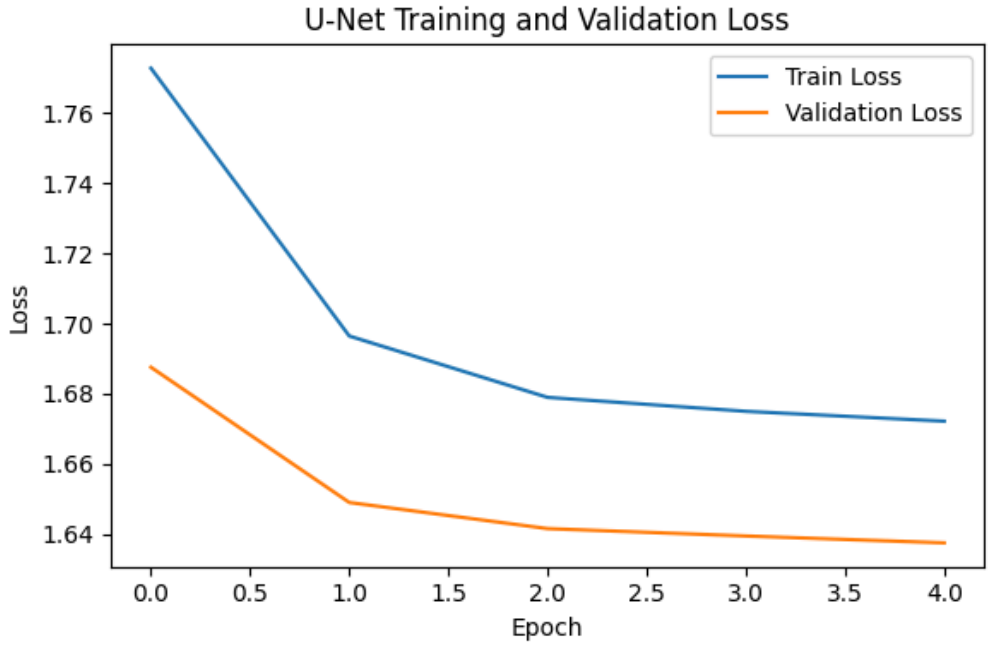



**Fig 4**. The loss curves for the training and validation subsets for our U-Net model under baseline preprocessing conditions. Each loss decreases consistently as epochs increase with minimal divergence and no significant signs of overfitting.

Both the training and validation loss decreased gradually which indicates the model is learning well, even with the small subset of volumes selected. The validation loss closely follows the training loss. Nearly identical loss patterns were seen in both the intensity normalization and Gaussian smoothing preprocessing techniques, which indicates that the convergence behavior across all three conditions is comparable.

**TABLE 2.** Quantitative mean $\pm$ standard deviation performance metrics across each preprocessing strategy. Metrics computed across the slices (n = 80) evaluated from the test volume. Sensitivity, Positive Predictive Value (PPV), and Dice Similarity Coefficient (DSC) were used to determine accuracy of segmentation.

| **Preprocessing Method** | **Mean ± STD Dice Similarity Coefficient** | **Mean ± STD Positive Predictive Value** | **Mean ± STD Sensitivity** |
|---|---|---|---|
| Baseline (No preprocessing) | 0.8330 ± 0.1549 | 0.7696 ± 0.1719 | 0.9356 ± 0.1260 |
| Normalization | 0.8400 ± 0.1596 | 0.7859 ± 0.1891 | 0.9402 ± 0.0899 |
| Gaussian | 0.8238 ± 0.2180 | 0.8376 ± 0.2211 | 0.8159 ± 0.2257 |

As displayed in Table 2, all three preprocessing strategies provided very strong segmentation performance. All 3 cases had dice similarity scores that were > 0.80. Z-score normalization provided the highest DSC and Similarity scores amongst the three, showing its slightly improved detection and predicted overlap of tumor regions. Gaussian smoothing achieved the highest PPV metric by a wide margin, indicating it is making solid distinction while maintaining few false positives. The baseline model was still very competitive in metrics, falling just slightly behind the others.

Even though the mean performance metrics represented high performance in segmentation, there was a high degree of variability observed through the individual slice displayed visually. Through multiple cases it produced no segmentation outputs, which resulted in metrics of 0.000. This could be due to either no tumor in the slice at all or the model completely missing the tumor. An example of a failed case is displayed here:

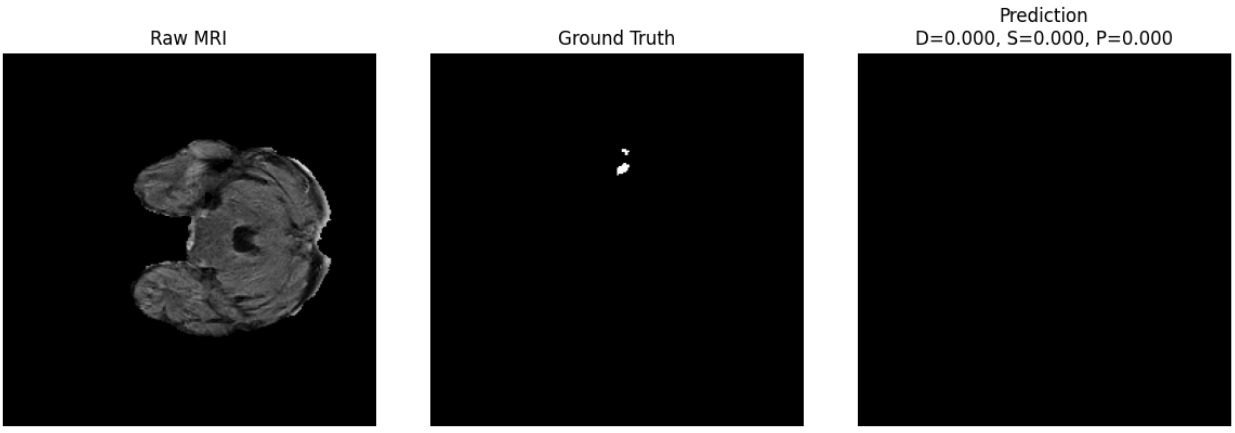



**Fig 5.** Raw MRI image (left), ground truth (center), and CNN-predicted tumor mask (right) on an example slice using baseline (none) preprocessing. The model failed to detect any tumor region at all, which produced metrics of 0.000 across all the quantitative measurements. This highlights a large degree of variability between slices.

With these inconsistencies in some of the individual slices like what is shown in Figure 5, the model still produced extremely well in total across the entire single volume test subset. There is a clear limitation here with the performance being heavily reliant on incorporating the entire slice set. Using more focused data augmentation, this can likely be amended to help improve consistent visual results across example cases. Increasing the amount of training, validation, and testing data or increasing the epoch count could also provide a lot of benefit in helping the model learn more to catch any potentially missed tumor pixels. One exemplary successful segmentation output utilizing the Gaussian preprocessing is shown below.

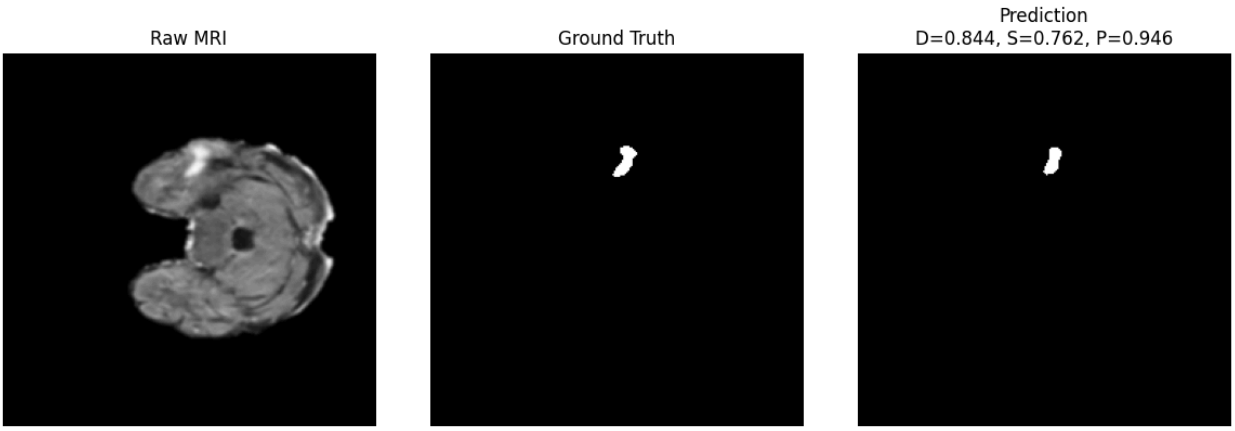



**Fig 6.** Raw MRI image (left), ground truth (center) and CNN-predicted tumor mask (right) on an example slice using Gaussian preprocessing. Highest individual performance in terms of visual performance across all three preprocessing methods. Quantitative Metrics: Dice Similarity Coefficient = 0.844, Sensitivity = 0.762, Positive Predictive Value = 0.946.

As showcased in Figure 6, the U-Net model is able to perform well at capturing the region of tumor in correspondence to the ground truth mask. Both the structure and positioning and the predicted tumor region is relatively intact, as shown by the high DSC score. In addition to the raw masks, we obtained overlay images incorporating the raw MRI images. The high performing prediction from Figure 6 is further expanded upon in Figure 7, which appears more visually interpretable.

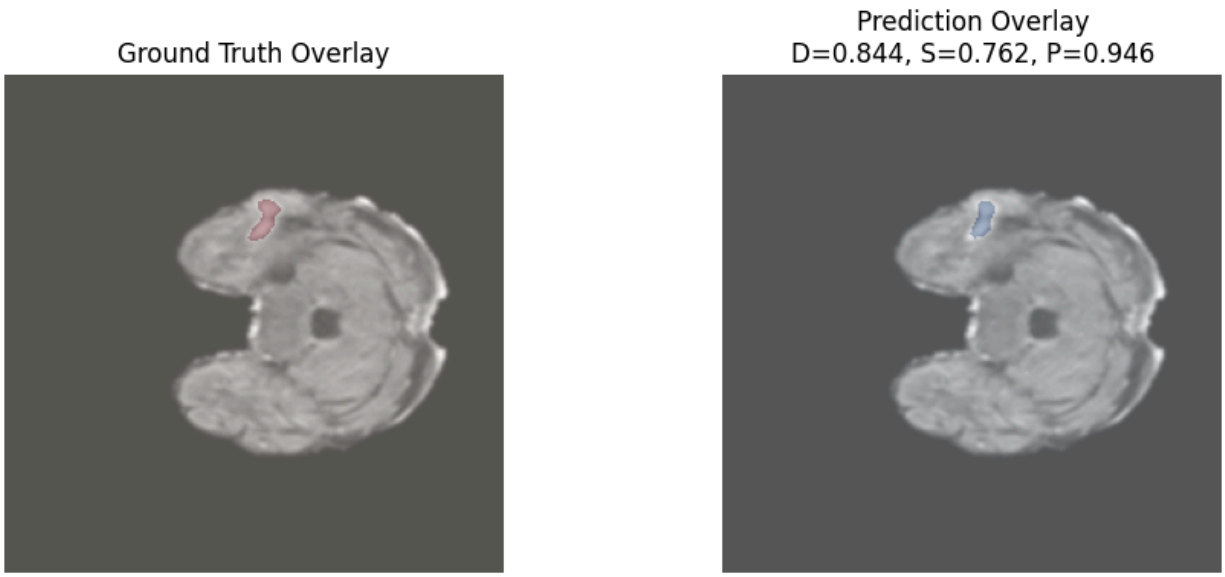



**Fig 7.** Visual overlay of CNN-model segmentation produced ground truth (left) and predicted segmentation (right) layered onto the Raw MRI slice. Displaying the highest performance through Gaussian preprocessing as indicated in Figure 6. The prediction closely resembles the ground truth in structure and location, substantiated by the high DSC score (0.844).

In total, the CNN approach showed much more effective improvement over the threshold baseline approach. It yields much higher quantitative metrics and in some of the cases high performing visual products as well. This CNN model needed roughly 15 minutes for training and evaluation on the 5 selected volumes, which signals that these more complex methods require significantly more computational cost. Given the improved performance and potential to reduce this, this runtime seems completely reasonable. The overall metrics signal that we observed some effective segmentation, but there is room for improvement in consistent visual outputs.

## Discussion

### I. Comparison to Pereira's Work

The results here signal strong segmentation performance, with Dice Similarity Coefficients surpassing 0.80 across all three preprocessing techniques (Baseline = 0.8330, Gaussian = 0.8238, Normalization = 0.8400). This is consistent with Pereira et al's. work, showing the effectiveness of utilizing convolutional neural networks for brain tumor segmentation. Pereira et al. achieved Dice scores as high as 0.88 when working with complete tumor regions on the BraTS 2013 challenge dataset.

Although our results in this study were less impressive than the peak that was reported in the original work, they still remain comparable even despite some key differences in methodology utilized. Their work was done on multi-class segmentation across various tumor subregions, whereas our implementation focused only on binary whole-tumor segmentation. The original study also employed a substantially larger dataset and utilized thousands of patches through extensive data augmentation. Our project was limited to only 5 volumes due to computational constraints, with the test set including 80 test slices.

There are some key distinctions in training strategy that can also contribute to the difference in performance. Pereira et al. used deeper CNN architectures, patch-based learning as opposed to the 2D slice-based approach, and much longer training regiments for their models. The implementation presented here employs only 5 training epochs and working in the 2D domain, yet still produced relatively impressive performances in segmentation ability. This indicates that meaningful results can still be achieved when limited by computational and data conditions.

The Pereira study had achieved efficient segmentation and reported runtimes at an average of 8 minutes despite the complexity employed. The implementation here needed roughly ~15 minutes for training and validation on just a small subset of data. This is another key distinction that substantiates the need for optimized hardware and pipeline complexity. Given the constraints in the computational setup for the U-Net implemented here, the runtime remained reasonable for the results obtained.

## II. Insights

An initial insight to gain from this study is that segmentation through CNN-models vastly outperforms traditional methods like intensity-based thresholding. The thresholding-based method failed to generate any meaningful segmentation and produced vastly over-segmented results. This emphasizes that relying solely on pixel intensity values is not appropriate for identifying tumors and that learning subtle spatial features is crucial for segmentation tasks.

Another key discernment to take from this study is how preprocessing techniques influence model performance. Intensity normalization through z-score produced the highest Sensitivity and Dice Similarity Coefficient, indicating that through standardizing the distributions it improves the CNN-model's ability to successfully find tumor regions and more closely resemble the ground truth. On the contrary, Gaussian smoothing resulted in the highest Positive Predictive Value, which showcased improved precision. Employing the Gaussian technique in this implementation results in the lowest number of false positives, meaning that positive detection is highly reliable. These results signify that various preprocessing techniques have different strengths and can optimize certain aspects of segmentation performance. It also indicates that

there is no “ideal” preprocessing technique that can universally improve all metrics, but rather needs to be used on a use-case basis depending on the data type used and the problem you are trying to solve.

While the quantitative metrics produced were high, qualitatively looking at the slices revealed some severe failure cases and high degree of variability amongst them. Some of the slices showed impressive segmentation (>.85 DSC), while others produce 0.000 across all three metrics. This substantiates that a strong average performance can hide some of the instability at the individual slice level.

## III. Limitations

This study has a variety of major limitations. The most notable is the dataset size utilized, as only five volumes were used due to computational and time constraints. Three volumes were used in the training subset, with validation and testing only having one volume each. With limited volumes, especially in the training and validation, it alters the model’s ability to generalize well and increases the amount of variation between individual slices.

Another limitation to the study is the inherent constraints of using a 2D slice-based approach rather than an entire 3D volumetric method. Although this simplifies the computational cost, it prevents the model from learning contextual differences across adjacent slices. This is an important tradeoff because it will reduce the ability for the model to accurately capture tumor structure.

The length of training was relatively low which additionally impacts the model’s performance negatively. The model may not have time to fully converge and become plateaued in performance capability without assessing a more extensive training period. In the loss function graph observed, the model begins to have relatively little change at the end of the epoch cycle, indicating that it is approaching convergence but not necessarily reaching it yet.

The preprocessing techniques were implemented without any further optimization or combination, which could have restricted some potential improvements as far as performance goes.

Lastly, the variability in the slice performances substantiates that the CNN-system struggles with smaller tumor regions; which would lead to potential false negatives. This is especially important due to the high cost associated with false negatives in a brain tumor context.

## IV. Impact on Understanding

The results obtained in this study strengthen the understanding of using deep learning approaches to address the difficult problem of accurate and efficient brain tumor segmentation in MRI data. The CNN model both quantitatively and qualitatively proved its effectiveness in understanding subtle complexities and spatial characteristics in image analysis as opposed to the more traditional methods that are unable to do so without much more extensive feature engineering and tweaking.

Observing the results of model performance with both visualization and numerical metrics is essential to understanding how effective segmentation models work. The variability through the visual outputs especially reinforces this understanding. The findings also hint that while the preprocessing strategies that were employed did slightly improve performance, this is not a consistent pattern across all metrics of performance. Certain techniques may need to be leveraged to obtain what the desired outcome for the specific use-case is. If the goal is to improve precision, Gaussian smoothing would be the most ideal amongst the three techniques. However, if overlap accuracy is the preferred outcome, normalization would be more appropriate.

This study gives some key insights on how CNN models work under constraints and reinforces how important dataset size, quality, and training strategy all contribute to attaining consistent and reliable segmentation.

## Conclusion

This study showcases a U-Net CNN architecture that is able to produce strong brain tumor segmentation (DSC > 0.80 mean) utilizing MRI data. Intensity normalization was the most effective in improving overlap and general detection, whereas Gaussian smoothing improves precision. Different preprocessing techniques contribute to different aspects of performance. In spite of limitations like a smaller dataset size, simplified 2D slice-based approach, and lack of a prolonged training period; the results remained comparable to the original work done by Pereira et al. The variability in slice outputs substantiates the need for more data augmentation and larger datasets, which requires more computational power. In total, the conclusions back how effective deep learning can be for segmentation in medical imaging.

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